

DEEP-TECHNOLOGY RESEARCH

φ Coherent Inc.

Designs grounded in φ coherent mathematics.

ONTARIO, CANADA · PRE-SEED ROUND · 2026

THE FOUNDATION

One identity.

Every system we build derives from it.

Zero free parameters.

When every constant, every dimension, every transform is constrained by a
single mathematical identity,
systems stop fighting themselves. They compose.

PHI-COHERENT INC. — DESIGNS GROUNDED IN ϕ COHERENT MATHEMATICS.

WHAT WE BUILD

Four disciplines. One foundation.

PILLAR I

Quantum Computing

Production-grade quantum simulation and a novel family of quantum error correction codes. Benchmarked against published noise specifications for 5 devices: IBM Heron R2, IBM Eagle R3, Google Willow, IonQ Forte, and Quantinuum H2.

PILLAR II

Compression

A complete audio · image · video codec family. Lossless by default, deterministic across architectures. Already outperforms FLAC on all lossless metrics. Core to the AGI sensory pipeline.

PILLAR III

Artificial General Intelligence

A multi-sensory, multi-cognitive architecture derived entirely from first principles. Zero free parameters. Every instance born with a cryptographic quantum-resistant identity.

PILLAR IV

Cryptography & Currency

A post-quantum cryptographic standard and ϕ Currency protocol. Hash-based throughout — no algebraic hardness assumptions. Quantum-resistant by construction, not by patch.

Error correction that wins where surface codes fail.

$p > 1\%$

Functional above the surface-code threshold

Surface codes fail ($p_L \rightarrow 50\%$) beyond $\sim 1\%$ physical error rate. Our multi-stratum codes remain functional at $p = 30\%$, $p_L \approx 0.087$.

$\leq 1.6 \text{ mHa}$

VQE chemical accuracy on H_2

UCCSD ansatz, noise-free lower bound. LiH error: 0.034 mHa vs FCI reference (-7.8825 Ha).

5

Published hardware noise models

IBM Heron R2 · IBM Eagle R3 · Google Willow · IonQ Forte · Quantinuum H2 — not real-device measurements.

78

Passing tests, fully reproducible

Wilson 95% CI on all logical error rates. Stim cross-validated. Snapshot-pinned, regenerated from source in minutes.

$94\times$

Lower logical error rate — 4-level multi-stratum ensemble vs best matched single surface code

530-qubit 4-level ensemble vs $d \leq 16$ surface code (511 qubits) at $p = 0.001$. Circuit-level depolarizing simulation, 15,000 trajectories.

Header-only C++

Density-matrix Kraus noise

Monte Carlo trajectories

DEJMPS entanglement distillation

φ -spaced phase estimation

Cross-stratum veto

Stim cross-validated

Shor N=15 validated

A complete codec family. Built to last.

Audio

Lossless: beats FLAC in compression ratio, encode speed, and decode speed across all tested material. Lossy levels in active development.

COMPLETE

Image

21×34 golden-rectangle block geometry. Lossless by default. Cross-architecture byte-identical output. Full test coverage 1×1 to 2048×2048.

COMPLETE

Video

ϕ Coherent GOP (21 frames), golden-angle spiral motion estimation, ϕ Interpolated sub-pixel compensation. Round-trip deterministic.

IN PROGRESS

Base

Lossless general-purpose codec. Beats zlib on all RPC payload classes. Blockchain-deterministic. ~45% bandwidth reduction on crypto sync traffic.

COMPLETE

Why codecs are central: They are the sensory pipeline for the AGI architecture. Completion of the audio and video codecs directly enables the full HEAR and SEE perceptual front-end — gating the integrated ϕ Engine deployment.

ϕ Engine: a mind derived, not designed.

SENSE · PERCEIVE

Feel — somatic awareness

Smell — chemical signal

Taste — gustatory input

Hear — ϕ Audio codec front-end

See — ϕ Visual codec front-end

Fuse — multimodal integration

THINK · COGNITION

Language — text encoding

Reasoning — logical inference

Memory — episodic store

World Model — environment state

Metacognition — self-monitoring

Theory of Mind

PROPERTIES

Zero free parameters

273-byte universal embedding

Post-quantum identity at birth

Signed checkpoints + outputs

GPU-accelerated (CUDA/Metal/Vulkan)

376 tests across all modules

Every weight, every dimension, every constant derives from a single mathematical identity. No empirical tuning. No hyperparameter search. The architecture is not discovered — it is derived.

THE PLATFORM · 8 METHODS · 10 DESIGNS

Deep IP. Built in parallel.

EIGHT METHODS

- ϕ Crypt — unified cryptographic standard
- Signal decomposition framework
- Coherent AI architecture
- Quantum computing framework
- Cellular framework — mathematical substrate for all systems
- Materials hierarchy — composition derived from atomic structure
- Microstructure design
- Impeller fluid dynamics

FOUR SOFTWARE DESIGNS

- ϕ Coherent codec family — audio · image · video · base
- ϕ Crypt reference implementation + OpenSSL provider
- Quantum computing stack — 23 packages, 7 functional layers
- Coherent AI models — ϕ Engine sensory + cognitive suite

SIX HARDWARE DESIGNS

- Bearings · Gyroscope · Impeller
- Centrifugal pump · Ram pump · Nozzle

License structure: All IP owned by Phi-Coherent Inc. Published under AGPLv3 open-source with commercial licensing available. Revenue flows to Phi-Coherent Inc.

THE OPPORTUNITY

Four converging markets. One coherent platform.

\$450B+

Quantum Computing

Global market by 2035 · Error correction is the critical unsolved bottleneck

\$1T+

AI / AGI Infrastructure

Projected by 2032 · Foundation model compute demands coherent, efficient architectures

\$15B+

Codec Licensing

Annual TAM · Streaming, broadcast, edge-AI, and satellite demand better compression

\$7B+

Post-Quantum Cryptography

By 2030 · NIST PQC transition underway · Hash-based standards gaining mandate

PRE-SEED RAISE · ONTARIO, CANADA

\$2M – \$5M

USE OF FUNDS

- 1 **Research team** — 3 to 5 PhD researchers spanning electrical, mechanical, chemical, quantum, and applied mathematics to advance patents and experimental proof
- 2 **Research acceleration** — lossy codec completion, full AGI sensory integration, quantum hardware validation partnerships
- 3 **Patent prosecution** — full protection of the ϕ Coherent method and design IP portfolio across key jurisdictions
- 4 **Licensing go-to-market** — commercial licensing of methods and designs across codec, cryptography, quantum, AI, and hardware verticals

CANADIAN GOVERNMENT PATHWAY

NRC-IRAP — Industrial R&D funding eligible

NSERC Alliance — Academic partnership-ready

SR&ED / Ontario ITC — Qualified R&D expenditures

SDTC — Sustainable tech + advanced manufacturing tracks

REACH OUT

jbadwal@phicoherent.com

Ontario, Canada · phicoherent.com